

Computing Reference Notes

Python basics

Variables store values, conditions choose paths, loops repeat work, functions group reusable logic, and lists hold ordered collections.

In Git, a branch isolates work. Commits record logical changes. Merge conflicts happen when Git can't combine overlapping edits automatically.

SQL and APIs

A SQL JOIN combines related rows using a condition. GROUP BY collects rows so aggregates such as COUNT or SUM can be calculated.

REST APIs expose routes that accept HTTP requests. Request validation should happen before business logic where practical.